Programmable Phase and Amplitude Control in Both Illumination and Detection Paths of Fluorescence Microscopes

A technical review of what exists, what the physics permits, and how to build one

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0. Short answer

Yes — such systems exist and have been built, but almost none of them do what the question literally asks, and the gap between "what exists" and "what the question describes" is itself the most interesting part of the field.

Three separate statements need to be kept apart:

1. **Systems with programmable wavefront control in *both* paths simultaneously exist and are mature.** The flagship examples are Betzig's AO lattice light-sheet microscope (2018) and its successor MOSAIC (Nature Methods, 2026). Both carry an independent adaptive-optics channel in the excitation arm and another in the detection arm.
2. **Their control is *not* "across the whole field of view."** It is *piecewise isoplanatic*: a single pupil-conjugate modulator applies one shift-invariant complex pupil filter, valid only over the local isoplanatic patch (~10–100 μm in tissue). Large volumes are covered by tiling sub-volumes, each with its own excitation and detection correction, acquired sequentially.
3. **True full-field, space-variant complex control on the detection side is fundamentally harder than on the excitation side**, and for incoherent fluorescence it is partly forbidden by physics rather than merely unbuilt. This is the real frontier.

There is a fourth statement, which the differentiable-optics literature has made unavoidable:

4. **A large part of what these instruments do optically can be moved into computation** — aberration correction, optical sectioning, axial scanning, spectral splitting — and increasingly has been. But the exchange rate is not uniform. Computation replaces optics that *correct*; it cannot replace optics that *collect*. Photons, the pupil cutoff and the modulator's space-bandwidth product are hard floors that no reconstruction crosses.

The rest of this document works through why, catalogues the systems that exist, assesses how much of the simulated design literature has reached hardware, and ends with a concrete build architecture.

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1. The physics that constrains every design

Before cataloguing hardware, it is worth being precise about what a modulator can and cannot do in each path, because most of the confusion in this area comes from treating the two arms symmetrically. They are not symmetric.

1.1 Excitation is coherent; detection is not

The excitation beam is a single coherent laser field. A phase-only spatial light modulator in the excitation pupil can therefore synthesise essentially *any* physically realisable 3D intensity distribution in the sample within the numerical-aperture and space-bandwidth budget: optical lattices, Bessel beams, doughnuts, speckle, arbitrary holographic spot arrays, per-spot pre-compensated aberrations. Full complex control is meaningful and achievable.

Fluorescence emission is spatially incoherent. Each emitter is coherent *with itself* but mutually incoherent with every other emitter. The consequences are severe:

- You **cannot phase-conjugate the emission field** of an extended labelled object. There is no single field to conjugate; the mutual coherence function is a delta function at the object plane.
- A modulator in the **detection pupil** implements a *shift-invariant complex pupil filter* — it engineers the PSF. This is powerful (see §3.3) but it applies one and the same operation to every point in the field. It is not per-point control.
- A modulator in a **detection image plane** implements a *space-variant amplitude mask*. Phase applied at an image plane to incoherent light is invisible at the detector (intensity is unchanged), so an image-plane device is an aperture selector, not a phase controller. This is exactly the programmable array microscope.
- The one exception: **self-interference holography**. A single point emitter is coherent with itself, so if you split its wavefront and re-interfere the two copies, you get a fringe pattern encoding its 3D position. This is what FINCH exploits, and it is the only route to genuine complex-field encoding of an incoherent object on the detection side.

1.2 Isoplanatism is the real limit on "whole field of view"

A pupil-plane corrector inverts one wavefront error. Sample-induced aberrations are field-dependent (anisoplanatic). The corrected region is the isoplanatic patch — in biological tissue typically on the order of 10–100 μm laterally, shrinking with numerical aperture and depth.

A recent quantitative treatment of this problem (MIT, 2025) measured that in a strongly aberrated setup, correcting a single isoplanatic patch left roughly 20% of the addressable field at moderate quality and only about 2% at the Maréchal diffraction-limited criterion; more than 50 separate patch corrections would have been needed to cover the field serially. That is the honest scale of the problem.

Four architectures attack it:

Approach	Mechanism	Cost
Serial tiling	Re-measure and re-correct per sub-volume	Time; incompatible with fast dynamics
Multi-conjugate AO (MCAO)	Second corrector conjugate to the aberrating layer, not the pupil	Complexity; assumes layered aberration
Pupil segmentation / multi-pupil AO	Tile the pupil, each sub-pupil serves a different field region	Loss of NA per channel
Aberration-space holography	Merge per-site propagation kernels into one hologram	Excitation/projection only, so far

1.3 Complex (amplitude *and* phase) modulation on a phase-only device

Phase-only LCoS SLMs are the workhorse. Amplitude control is obtained by encoding — double-phase holograms, Lee holograms, superpixel methods, or diffracting unwanted energy out of the pupil. Each costs space-bandwidth product and diffraction efficiency.

On the **excitation** side this is fine: you have laser power to spare.

On the **detection** side it is usually a mistake. Amplitude modulation of fluorescence throws away photons that are already scarce, and LCoS devices are polarisation-selective (≈50% loss on unpolarised emission), pixelated, and chromatically dispersive when used as blazed gratings across a broad emission band. This is precisely why Betzig's group used a **deformable mirror** rather than an SLM in the detection arm — achromatic, polarisation-insensitive, near-unity reflectivity — accepting far fewer degrees of freedom in exchange for photon efficiency.

Design rule that falls out of this: *SLM in excitation, DM in detection*, unless you are doing PSF engineering or self-interference holography, in which cases an SLM in detection earns its losses.

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2. Taxonomy: where the modulator sits and what you get

Path	Conjugate plane	Device	Function	Resulting modality
Excitation	Pupil (Fourier)	LCoS SLM / DM	Wavefront, beam shaping	AO, lattice/Bessel LS, holographic spots, speckle
Excitation	Image	DMD / SLM	Intensity pattern	SIM, HiLo, multifocal, targeted illumination
Detection	Pupil (Fourier)	SLM / DM / phase mask	Complex pupil filter	PSF engineering (DH, Tetrapod), AO, multifocus
Detection	Image	DMD / SLM	Space-variant amplitude	Programmable array microscope, confocal masking
Detection	Pupil, self-interfering	SLM / birefringent lens	Self-interference hologram	FINCH, COACH, I-COACH
Both	Shared conjugate	Single DMD	Simultaneous patterning of both	PAM (aperture correlation)
Both	Independent	SLM + DM	Independent AO per arm	**AO-LLSM, MOSAIC**

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3. Systems that have actually been built

3.1 AO lattice light-sheet microscopy — the canonical dual-path instrument

Liu, Upadhyayula, Milkie, Betzig et al., Science, 2018.

Architecture:

- Excitation arm: an SLM shapes the beam into an optical lattice (an array of coherently interfering non-diffracting beams), producing a sheet roughly 0.7 μm thick over ~15 μm of propagation. The *same* SLM adds the excitation-path aberration correction.
- Detection arm: an independent deformable mirror in the detection pupil corrects emission-path aberrations.
- Guide star: a two-photon-excited fluorescent spot created inside the specimen. It can be created through either arm, so each arm's aberration is measured through the path it actually uses. Scanning the guide star over a region yields an *averaged* correction for that region, which is more robust than a single-point correction and spreads photon load.
- Correction time: ~70 ms, fast enough to interleave with live imaging.
- Large volumes are assembled from sub-volumes, each with independent excitation and detection corrections.

The system is effectively three microscopes on one table — the published parts list runs to dozens of lenses, mirrors and galvos, two lasers, three objectives and four cameras.

Key point for the question as posed: this is dual-path programmable phase control, but the control is *one wavefront per sub-volume per arm*, not per-field-point.

3.2 MOSAIC — the current state of the art

Fu, Liu, Milkie, Ruan et al. (Upadhyayula / Betzig / Legant), Nature Methods, 2026.

A reconfigurable ~1 m^3 instrument that switches between more than ten modalities — oblique transmitted-light, widefield epifluorescence, 3D-SIM, lattice light-sheet, LLS-SIM, image scanning microscopy, and two-photon point scanning — with **adaptive-optics wavefront sensing and correction in both excitation and emission paths for all modes**. Three objectives in a temperature-controlled chamber; a 69-actuator deformable mirror.

Demonstrated: single-molecule tracking in live adherent cells; ~1 mm × 0.75 mm fields of cells volumetrically imaged at roughly 260 × 260 × 431 nm over 24 hours; single-spine functional imaging in mouse cortex; millimetre-scale mapping in cleared tissue.

This is the closest existing answer to the question. It is not, however, full-FOV complex control — it is best-in-class isoplanatic-patch AO in both arms, made practical across many modalities.

3.3 Detection-side PSF engineering

A phase mask (SLM or fabricated fused silica) in a 4f relay at the detection Fourier plane converts depth into a distinctive lateral PSF signature, giving volumetric single-molecule localisation from a 2D camera:

- **Double-helix PSF** (Piestun / Moerner) — two lobes rotating with z.
- **Tetrapod PSF** (Shechtman, Weiss, Backer, Sahl, Moerner) — optimised for very large axial ranges, used for multi-particle 3D tracking.
- **Learned PSFs** — DeepSTORM3D (Nehme, Michaeli, Shechtman, Nat Methods 2020) trains a CNN to localise dense overlapping emitters *and* to design the optimal phase mask for that task. The optics and the estimator are co-designed.
- **Field-dependent PSF calibration** (Science Advances, 2024) — engineered PSFs vary across the field; the fix was to measure PSFs at a grid of field positions and interpolate, enabling large-FOV 3D localisation. Note this is *computational* accommodation of anisoplanatism, not optical correction of it.

3.4 Self-interference holography — FINCH and its descendants

Rosen & Brooker, 2007–2008; FINCHSCOPE. An SLM in the emission path displays a multiplexed diffractive lens that splits each emitter's wavefront into two spherical waves with different curvature. They self-interfere at the camera. Three phase-shifted exposures (0, 2π/3, 4π/3) are combined into one complex hologram, which is numerically propagated to reconstruct any plane — motionless, non-scanning 3D from a single complex hologram.

Notable properties:

- **Super-resolution of roughly 1.5× beyond the equivalent Rayleigh limit**, attributed to violation of the Lagrange invariant in this configuration.
- Later implementations replaced the lossy pixelated SLM: transmission liquid-crystal GRIN lenses, and **birefringent α-BBO / calcite crystal interferometers** (Nature Photonics, 2016) which reached a 149 nm lateral PSF at 590 nm with high-NA oil immersion, and resolved three-colour Golgi labelling in HeLa cells.
- Single-shot variants (FISCH; multiplexed sub-aperture FINCH) remove the three-exposure penalty.
- **FINCH + structured illumination** has been demonstrated, stacking illumination-side and detection-side resolution gains.
- **COACH / I-COACH** (Rosen) generalise the emission-path mask from a diffractive lens to a coded/chaotic phase mask, with reconstruction by correlation against a calibrated point response.

FINCH is the clearest existing example of programmable *complex* control of the detection light. Its resolution gain is modest and its photon efficiency poor, which is why it has not displaced conventional detection.

3.5 Simultaneous modulation of both paths with one device

The **programmable array microscope (PAM)** (Hanley, Verveer, Jovin) places a DMD or LCD at a plane conjugate to the specimen that is shared by illumination and detection. The same pixel pattern illuminates the sample and gates the returning light — programmable multipoint confocal with arbitrary, computer-controlled aperture patterns, faster and more photon-efficient than a scanning confocal, plus the conjugate (rejected) image as a bonus channel.

Amplitude only, and the shared-plane trick works precisely because it does *not* need phase control on the detection side. This whole family — programmable pinholes, aperture correlation, Hadamard coding, image scanning — is treated in full in §4.

3.6 SIM with adaptive optics — dual-path in the super-resolution regime

- **AO 2P-ISIM** (Zheng, Wu, Shroff et al., Nat Methods 2017): nonlinear guide star, deformable mirror imaged onto the objective back focal plane, and an emission scanner that either rescans the fluorescence for super-resolution or descans it onto a Shack–Hartmann sensor for wavefront measurement. Up to 40× peak-intensity gain; 176 nm lateral / 729 nm axial recovered at ~250 μm depth.
- **AO-SIM in vivo** (Turcotte, Liang, ... Betzig, Ji, PNAS 2019): AO plus acquisition/reconstruction changes to survive sample motion, giving nanoscale imaging of dendrites and spines in live zebrafish larvae and mouse brain.
- **AO-OS-SIM** (Li, Zhang, ... Ji, Science Advances 2020): optical-sectioning SIM with AO for fast widefield structural and functional brain imaging.
- **AO 2P-MSIM** (PhotoniX, 2023): adaptive-optical two-photon multifocal SIM for deep brain slices and in vivo zebrafish motoneurons.

3.7 Illumination-side randomisation: RIM

Random illumination microscopy (Mangeat, Labouesse, Idier, Sentenac et al., Cell Reports Methods 2021) sidesteps the excitation-side aberration problem entirely. Rather than controlling the illumination precisely, it uses ~100 uncontrolled speckle patterns (generated with an SLM or diffuser) and reconstructs from the *variance* of the resulting images.

- Resolution matching 3D-SIM (≈2×), proven identifiable from variance matching in the noiseless case when the speckle correlation length matches the observation PSF width.
- **Unaffected by excitation-path aberrations** — you do not need to know the pattern, so distortion of it is harmless. This is why it works deep inside scattering tissue where SIM's known-pattern assumption breaks.
- **EDF-RIM** (Light: Science & Applications, 2024) adds extended depth of field for an order-of-magnitude speed and dose improvement on projective imaging of thick tissue.

RIM is the interesting counter-argument to the whole premise: sometimes *less* illumination control is better.

3.8 Computational dual-path correction: the incoherent response matrix

Lim, Kang, Kim, Choi & Yoon (Korea University / IBS / Pusan), 2024–2025. Replace the point detector of a laser-scanning microscope with a camera and record an **incoherent response matrix** — the full camera image for every scan position. A **dual deconvolution** algorithm decomposes the IRM into the excitation OTF, the emission OTF, and the object spectrum, correcting excitation and emission PSFs *independently and computationally*, with essentially no added hardware.

Result: λ/4 lateral resolution — 130 nm at 180 μm depth in mouse brain tissue via two-photon "virtual" SIM. Spatially varying aberrations are handled by dividing the field into subregions (7×7 patches of ~20 μm across a 145 × 145 μm ROI) and solving each separately.

This is the most direct computational analogue of the dual-path hardware systems, and it makes the honest point that the excitation/emission split can be recovered in software from richer measurements.

3.9 Extending the corrected field

- **Multi-pupil adaptive optics** (Park, Kong, Zhou & Cui, Nat Methods 2017): tile the pupil into sub-pupils, each carrying an independent wavefront for a different field region. 450 × 450 μm corrected — 9× the conventional area — and because the sub-wavefronts are independent, the system can image a *non-planar* surface (following vasculature at different depths simultaneously).
- **Conjugate AO** (Mertz group, Applied Optics 2015; Cui at Janelia): put the corrector conjugate to the aberrating layer rather than the pupil. Substantial FOV advantage when aberrations are layer-localised — demonstrated through mouse skull.
- **MCAO in microscopy**: modelled by Simmonds & Booth (2013); closed-loop confocal MCAO (Booth group, 2020); **plug-and-play deformable *lenses*** in the detection path with spinning sub-pupil wavefront measurement (Furieri, Bonora, Bassi, J. Biophotonics 2023) — a compact transmissive large-FOV corrector

for widefield and light-sheet detection arms.

- **Aberration-space holography** (Christen, Panuski, Propson & Englund, MIT, 2025): the most direct attack on the premise of the question. Instead of one global correction, individual propagation kernels — each carrying site-specific Zernike coefficients — are *summed into a single SLM hologram*. Reported: full-field anisoplanatic compensation with mean relative Strehl 0.78 versus 0.02-of-field diffraction-limited for the best isoplanatic correction; 8× larger diffraction-limited field for a 2D tweezer array; 12× addressable volume for a two-photon volumetric display in quantum dots. Correction was distilled by SVD onto 8 "principal isoplanatic modes." Open source as `slmsuite`.

  This is projection/excitation-side, and demonstrated on tweezers and a display rather than a microscope — but it is the template for what full-field excitation control looks like.

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4. Digital confocal: how a DMD or SLM replaces the physical pinhole

The programmable array microscope was mentioned in §3.5 as a curiosity. It deserves its own treatment, because "digital pinholing" is the one place where spatial light modulation in the *detection* path is unambiguously practical, mature and commercially deployed — and understanding exactly why illuminates the coherence argument in §1.1.

4.1 The confocal principle restated as a filtering operation

A confocal microscope is the product of two spatially confined functions:

```
h_eff(r) = h_exc(r) · [ h_det(r) ⊛ P(r) ]
```

where `h_exc` is the excitation PSF, `h_det` the detection PSF, and `P` the pinhole transmission. The pinhole is nothing more than an **amplitude mask at a plane conjugate to the object**. Two consequences follow immediately:

1. **Nothing about it requires a hole.** Any device that can apply a spatially selective transmission at a conjugate image plane implements a pinhole. A DMD micromirror flipped "on" is a pinhole; flipped "off" it is opaque.
2. **Nothing about it requires the mask to be physical.** If you image the whole conjugate plane onto a camera instead, you can apply `P(r)` in software after the fact. This is the "virtual" or "digital" pinhole.

The key point for this document: pinholing is **amplitude-only and image-plane**, which is precisely the operation §1.1 says *is* available for incoherent fluorescence. Phase at an image plane is invisible to an intensity detector, so nothing is lost by using a binary amplitude device. This is why the digital-pinhole family works so well while true complex detection-path control does not.

4.2 Three distinct meanings of "digital confocal"

These are routinely conflated. They are different physics with different photon budgets.

	Where the aperture acts	Rejects light	Photon cost	Examples
(a) Optical pinholing	Physically, at a conjugate image plane, before the detector	Yes — out-of-focus photons never reach the sensor	Rejected photons are lost (or captured as a second channel)	PAM, DMD confocal, spinning disk, Aurox Clarity
(b) Computational pinholing	In software, on a camera image	No — everything is detected, then weighted	No optical loss; but shot noise from rejected light is already baked in	MSIM digital pinhole, virtual-pinhole CLSM, Hadamard software aperture, ISM/Airyscan
(c) Demodulation sectioning	No aperture at all; contrast comes from illumination modulation	Statistically	Requires ≥2–3 frames per section	SIM sectioning, HiLo, aperture correlation, Hadamard

Approach (a) protects the detector from saturation and works with single-element detectors, but discards photons. Approach (b) keeps every photon and permits post-hoc choice of pinhole size, shape and position — but the noise from out-of-focus light is already in the measurement, so it improves contrast rather than SNR. Approach (c) is the most light-efficient in principle and is what gives structured illumination its optical sectioning.

Most real "digital confocal" instruments are hybrids. The PAM is (a) plus (c). Airyscan is (b) plus reassignment.

4.3 The programmable array microscope in detail

Verveer, Hanley, Verbeek, van Vliet & Jovin, J. Microsc. 189, 192 (1998) — theory.
Hanley, Verveer, Gemkow, Arndt-Jovin & Jovin, J. Microsc. 196, 317 (1999) — DMD implementation.
Heintzmann, Hanley, Arndt-Jovin & Jovin, J. Microsc. 204, 119 (2001) — dual-path version.

The defining feature of a PAM is a spatial light modulator sitting *in an image plane* — used either singly, or as a matched pair, for both illumination and detection.

Single-device (double-pass) architecture. One DMD serves both duties. Micromirrors in the "on" state pass excitation light to the sample; the returning fluorescence lands back on the same mirrors, and only light that arrives at an "on" mirror is reflected toward the camera. Illumination aperture and detection aperture are automatically, perfectly registered — no alignment drift is possible between them, which is the single biggest practical advantage over a two-mask system.

Conjugate and non-conjugate images. A DMD deflects light into two directions rather than absorbing it. Light hitting "on" mirrors forms the **conjugate image** `I_c` (the confocal channel); light hitting "off" mirrors forms the **non-conjugate image** `I_nc` (predominantly out-of-focus). A dual-path PAM records both simultaneously on two cameras, or two halves of one sensor. Two useful identities hold:

- For a *scanned aperture* pattern (a sparse grid stepped across the field): `I_c` is the confocal image and `I_nc` is essentially the complement, so `I_c + I_nc ≈` widefield.
- For a *pseudorandom* / *Hadamard* pattern with 50% duty cycle: `I_c` and `I_nc` are respectively the **sum** and the **difference** of a conventional and a confocal image. The confocal part must be retrieved by post-processing — but because half the field is illuminated at all times rather than a few percent, the signal level for a given integration time is dramatically higher.

That trade — sparse patterns give a clean section directly but waste light; dense pseudorandom patterns give far more signal but require demodulation — is the central design decision in every PAM-class instrument.

Programmability is the real payoff. Because the aperture pattern is software-defined, the same instrument can, without any hardware change:

- vary pinhole diameter and spacing per experiment, per wavelength, or per z-plane;
- switch between point, line and arbitrary aperture geometries (line apertures trade axial resolution for speed);
- confine illumination to an arbitrary region of interest, cutting photobleaching outside it;
- run pseudorandom Hadamard sequences for light efficiency, or sparse grids for maximum sectioning;
- reposition the effective pinhole electronically to correct lateral misregistration — auto-alignment with no moving parts.

Commercial and patent-level implementations extend this to **spectrally adaptive pinholes**: with a DMD placed at the entrance slit plane of a spectrometer, the aperture size can be synchronised to the detected spectral band so that the optical section thickness stays wavelength-independent across channels, and can be widened for dim fluorophores and narrowed for bright ones within the same acquisition.

4.4 Aperture correlation — the light-efficiency fix

Wilson, Juskaitis, Neil & Kozubek, Opt. Lett. 21, 1879 (1996); Juskaitis et al., Nature 383, 804 (1996).

A Nipkow-disk confocal must keep pinholes far apart to suppress crosstalk between neighbours, which typically leaves ~1% of the available light in use. Aperture correlation replaces the sparse pinhole array with a dense pattern (a grid or a binary grating) at ~50% duty cycle and recovers the section arithmetically:

```
I_section ∝ I_conjugate − I_non-conjugate
```

Both channels are collected simultaneously, so a single camera frame pair yields a section — real-time optical sectioning with an incandescent or LED source, no laser required. In the commercial Aurox Clarity implementation, a quartz disk carrying a reflective binary grating sits at a conjugate image plane, tilted slightly so that in-focus light passing through the grating apertures and defocused light reflected off the grating bars are imaged onto two halves of the same sensor, and a weighted subtraction yields the section.

Aperture correlation gives sectioning performance comparable to other methods without sacrificing lateral or axial resolution, using simple and inexpensive hardware; its limitation, shared with all structured-illumination sectioning, is signal-to-background in thick or strongly scattering specimens.

4.5 Hadamard microscopy — orthogonal codes instead of apertures

Parot, Sing-Long, Adam, Böhm, Fan, Farhi & Cohen, J. Phys. D 52, 144001 (2019).

A DMD projects, at neighbouring points in the sample, illumination sequences that are **orthogonal functions of time** drawn from a Hadamard matrix at 50% duty cycle, with a random binary mask inverting half the pixels to suppress crosstalk between repeated code locations. The camera stack is demodulated by matched filtering against the illumination code; a **software aperture** then keeps only the correlation signal spatially coincident with its own illumination point, and the correlation maps are summed into a section.

The important property for tissue: in a scattering sample, in-focus light is dispersed and mixed with out-of-focus background, but *scattered light loses its spatial correspondence with the code*. Matched filtering plus software aperture therefore rejects scattered and out-of-focus emission while preserving unscattered focal-plane light. Summing the raw frames instead of demodulating them returns the widefield image for free.

Compressed Hadamard imaging exploits the low-rank temporal structure of neuronal activity to reconstruct an optically sectioned movie at half the camera frame rate — matching HiLo's temporal resolution while retaining Hadamard's scattering rejection. Reported operating point: 24 patterns interleaved with their complements, DMD and camera updated at 66 Hz, reconstruction in 64 × 64 blocks with 40 principal components per block, over cortex-and-striatum-scale fields.

There is even a modulator-free variant: because certain Hadamard patterns repeat inside others, a **holographic metasurface** can in principle generate the whole set from one static pattern (demonstrated in simulation, 2025).

4.6 Image scanning microscopy — the digital pinhole taken to its limit

If the pinhole is replaced by a **detector array**, every pixel of the array is simultaneously its own small-pinhole confocal detector, viewing the sample from a slightly shifted position. The resulting 4D dataset is reconstructed by **pixel reassignment**: each sub-image is shifted toward the midpoint between illumination and detection spots before summing.

Quantitatively (Sheppard, Castello, Tortarolo, Deguchi, Koho, Vicidomini & Diaspro, JOSA A 37, 154 (2020)), for a detector array of two Airy units:

- effective PSF peak intensity **1.90×** that of widefield;
- transverse resolution **1.53×** better than widefield (peak can be raised a further 4% with an optimised reassignment factor);
- axial resolution 1.34× better than widefield with the standard factor — notably *better* than a confocal microscope with a pinhole the same size as the array, which manages only 1.04×;
- sectioning strength identical to a confocal with a pinhole equal to the array size;

- detectability of a point object in uniform background 1.6× higher than confocal.

This is the cleanest statement of what the digital pinhole buys: **confocal-class resolution and sectioning without confocal's photon rejection.** Zeiss Airyscan is the commercial embodiment (32-element GaAsP array; ~260 nm → 140–160 nm lateral, 600–700 nm → ~400 nm axial), and SPAD-array implementations retain the full 4D dataset for richer post-processing.

Related family members differ only in *where* the reassignment happens:

- **MSIM** (York, Shroff et al.) — DMD multifocal illumination, camera detection, then *digital* pinholing, ×½ contraction of each pinholed spot, summation, deconvolution.
- **iSIM** — the identical sequence performed *optically* with microlens arrays, matched physical pinholes and a galvo, hence at camera frame rate.
- **OPRA / re-scan confocal** — a galvo descans then rescans the emission at a different gain, performing the reassignment optically.

MSIM is worth dwelling on: steps I–VI of its pipeline are exactly a digital pinhole array applied post-hoc to camera frames. It is a pure software confocal riding on a DMD-structured illumination.

4.7 Practical engineering notes for a DMD detection arm

If you build one, these are the things that bite:

- **A DMD is a blazed grating, not a mask.** Micromirrors tilt (typically ±12°) on the diagonal, and the periodic array diffracts. With broadband fluorescence this produces wavelength-dependent diffraction efficiency and angular dispersion. Fine for on/off aperture duty at modest NA; a real problem if you try to use the DMD for phase encoding on the emission side.
- **Binary amplitude only.** Greyscale comes from time-dithering, which couples aperture "size" to exposure timing.
- **Two output channels for free.** The "off" state is not absorption — put a second camera there and you get the non-conjugate image at no photon cost. Any PAM design that throws it away is wasting half its information.
- **LCoS alternative.** Higher fill factor and greyscale, but polarisation-selective (≈50% loss on unpolarised fluorescence) and far slower. For image-plane aperture duty, DMD wins on nearly every axis.
- **Registration is everything.** In a two-device design, illumination and detection masks must be co-registered to a fraction of a pinhole across the field, including any field distortion. The double-pass single-DMD architecture eliminates this problem by construction, at the cost of needing the dichroic between DMD and camera.
- **Timing.** DMD frame rate, camera exposure and laser gating must be hardware-synchronised; a mismatch of one DMD frame smears the aperture pattern and destroys the section. Patents in this area specify explicit non-overlapping illumination and detection windows (`t0–t1` illumination on, `t2–t3` detection on) to suppress excitation bleed-through.

- **Adaptive optics folds in naturally.** A DMD acting as a dynamic reflective pinhole array *and* a second camera watching the rejected light gives you a free wavefront-quality metric: a sensorless AO loop can maximise the ratio of pinholed to rejected signal. This has been demonstrated in DMD-based confocal systems with a deformable mirror providing both correction and scanning.

4.8 Where digital pinholing sits relative to the rest of this document

Property	**Physical pinhole**	**Digital pinhole (DMD/SLM)**	**Complex detection modulation (§3.3, §3.4)**
Plane	Image	Image	Pupil
Quantity controlled	Amplitude	Amplitude	Amplitude *and* phase
Space-variant?	No (single) / fixed array	**Yes, arbitrary**	No — shift-invariant
Valid for incoherent light?	Yes	Yes	Yes, but only as a PSF filter
Photon efficiency	Poor	Moderate to good	Poor
Maturity	Ubiquitous	Commercial	Research

Digital pinholing is the *only* fully space-variant, whole-field detection-path modulation that is both physically meaningful for fluorescence and photon-viable. It just happens to be restricted to amplitude. That restriction is not a technological gap — it is the coherence argument of §1.1 expressed in hardware.

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5. Catalogue of SLM/DMD-based imaging: built in metal versus simulated

"Built in metal" here means a working instrument has been demonstrated with real photons and real samples. "Proof of principle" means bench demonstration on test targets or in a non-microscopy context. "Simulated" means numerical demonstration only.

5.1 Illumination-path techniques

Technique	Modulator role	Status
Structured illumination (2D/3D SIM)	Grating, ferroelectric LCoS or DMD generates and phase-steps fringes	**Commercial** (Zeiss Elyra, GE OMX, Nikon N-SIM)
Optical-sectioning SIM / ApoTome	Single-frequency grid, 3-phase demodulation	**Commercial**
HiLo	One structured + one uniform frame	**Built**, widely replicated
Lattice light-sheet	SLM synthesises optical lattice; annular pupil mask	**Built**, disseminated; commercialised (3i, Zeiss)
Bessel / Airy beam light sheets	SLM or axicon-equivalent phase	**Built**
Field synthesis	Scanned 1D pattern reproduces a 2D lattice	**Built** (Betzig lab)
Multifocal illumination (MSIM)	DMD multifocal array	**Built**
Random illumination microscopy	SLM/diffuser generates speckle ensembles	**Built**, commercialised (France BioImaging PCS)
Speckle-saturated 3D super-resolution	SLM random phase + saturation nonlinearity	**Proof of principle**
Holographic optogenetic photostimulation (CGH, 3D-SHOT, temporal focusing)	SLM makes 3D spot ensembles	**Built**, widely used
Holographic optical tweezers	SLM spot arrays	**Commercial**
STED / RESOLFT depletion shaping	SLM or vortex plate makes doughnut / bottle beam; parallelised RESOLFT makes doughnut arrays	**Commercial** (STED); parallel RESOLFT **built**
MINFLUX / MINSTED beam shaping	Structured excitation with a zero	**Commercial**
Two-photon temporal focusing	SLM + grating	**Built**
Aberration-space holography	Per-site Zernike kernels merged into one hologram	**Built**, but for tweezers and a volumetric display — **not yet a microscope**
Adaptive illumination (learned power vs depth)	Not a modulator per se; illumination policy learned from sample geometry	**Built** (Waller lab)

5.2 Detection-path techniques

Technique	Modulator role	Status
Programmable array microscope	DMD/SLM at conjugate image plane as programmable aperture array	**Built** (1998–2001), influential
Aperture correlation	Dense grid + conjugate/non-conjugate subtraction	**Commercial** (Aurox Clarity, Zeiss VivaTome)
DMD dynamic pinhole array + AO	DMD apertures, rejected light as AO metric	**Built** (TU Delft)
Programmable spectral pinhole	DMD at spectrometer slit; aperture synced to band	**Patented / built** (Zeiss)
PSF engineering (double-helix, tetrapod, self-bending, corkscrew)	Phase mask at detection pupil	**Commercial** (Double Helix Optics)
Deep-learning-designed PSFs (DeepSTORM3D)	Learned pupil phase on LC-SLM or etched silica	**Built**
Field-varying PSF calibration for large-FOV 3D SMLM	Fixed mask + interpolated PSF model	**Built**
Multifocus microscopy	Multifocus diffractive grating splits into z-planes	**Built**
FINCH / FINCHSCOPE	SLM as dual diffractive lens; self-interference hologram	**Built**; later crystal/GRIN versions
COACH / I-COACH	Coded phase mask + correlation reconstruction	**Built**
Detection-path AO (deformable mirror)	Pupil wavefront correction	**Built**, standard
Deformable-lens MCAO in detection	Transmissive plug-in large-FOV corrector	**Built** (Bonora/Bassi)
Adaptive/adjustable pinhole for AO calibration	Programmable square pinhole	**Built**
Cascaded (MPLC) space-variant detection correction	Multi-plane phase masks implement inverse transmission matrix	**Not built for fluorescence** — MPLC itself is built (10-plane, single SLM), imaging application is theoretical

5.3 Techniques spanning both paths

Technique	Status
PAM single-DMD double-pass (illumination + detection with one device)	**Built**
Dual-path PAM (simultaneous conjugate + non-conjugate)	**Built**
AO-LLSM (SLM excitation + DM detection, 2P guide star)	**Built**, disseminated
MOSAIC (AO in both arms, 12 modalities)	**Built** (2026)
AO 2P-ISIM (DM + emission rescan + Shack–Hartmann on descanned light)	**Built**
AO-SIM / AO-OS-SIM in vivo	**Built**
AO 2P-MSIM	**Built**
Dual deconvolution of the incoherent response matrix (computational correction of *both* excitation and emission PSFs)	**Built** (camera replaces PMT; algorithm published with code)
Simultaneous space-variant *complex* control in both arms	**Neither built nor simulated end to end** — the open problem of this document

5.4 Scattering, wavefront shaping and complex media

Technique	Status
Focusing through scattering media by wavefront shaping (Vellekoop–Mosk)	**Built**, foundational
Transmission-matrix measurement and inversion	**Built**
Digital optical phase conjugation (DOPC), TRUE/TROVE focusing	**Built**
F-SHARP and dynamic conjugate F-SHARP	**Built**
Multi-patch / simultaneous multi-point scattering compensation	**Built**
Multimode-fibre endoscopy (SLM pre-compensation of fibre transmission matrix)	**Built**; 2000 points at 1.44 kHz with a MEMS phase-only modulator
Distortion / reflection matrix microscopy	**Built** (label-free); fluorescence variants emerging
Image-guided computational holographic wavefront shaping	**Built** (2024)
Multi-pupil AO (segmented pupil, independent per-region wavefronts)	**Built**: 450 × 450 μm, non-planar imaging
Conjugate AO (corrector conjugate to aberrating layer)	**Built**, incl. through mouse skull
Multiconjugate AO in microscopy	**Modelled** (Simmonds & Booth 2013); **closed-loop confocal MCAO built**; general case still largely simulation

5.5 Computational imaging with modulators

Technique	Status
Single-pixel / compressive imaging with DMD	**Built**
Compressive confocal microscopy (random pinhole sets + single detector)	**Built**
Single-pixel PAM (SP-PAM)	**Built**
Single-pixel digital holography with DMD-encoded CGH patterns	**Built**
Ghost imaging	**Built**
Fourier ptychographic microscopy (LED array or DMD angle selection)	**Built**
Optical diffraction tomography with DMD illumination-angle scanning	**Built**
Light-field / Fourier light-field microscopy	**Built**
Hadamard / compressed Hadamard microscopy	**Built**
Metasurface-generated Hadamard patterns (modulator-free)	**Simulated only** (2025)
End-to-end differentiable co-design of optics + reconstruction network	**Built for cameras and for single-molecule PSFs**; for a full dual-path microscope, **simulated only**
Diffractive deep neural networks (D^2NN) as optical front ends	**Built for classification**; fluorescence microscopy use **simulated**
Spatial-mode demultiplexing (SPADE) superresolution via MPLC	**Proof of principle** on bench sources; fluorescence microscopy application **simulated**
Self-configuring universal linear optics meshes (Miller)	**Built on photonic chips**; not applied to a microscope

5.6 The hardware landscape, briefly

Device	Modulation	Speed	Efficiency	Where it belongs
Nematic LCoS SLM	Phase, greyscale, ~1–4 MPix	10–100 Hz	50% (polarisation) × diffraction	Excitation pupil
Ferroelectric LCoS	Binary phase	~1–5 kHz	Same polarisation penalty	SIM patterning
DMD	Binary amplitude (phase via binary holograms)	10–30 kHz	Broadband, polarisation-agnostic; diffractive	Image-plane apertures, SIM, Hadamard
TI phase-only MEMS light modulator (PLM)	4-bit piston phase, megapixel	kHz, ~10 kHz anticipated	Polarisation-agnostic, high	Emerging: fast wavefront shaping
Deformable mirror	Continuous phase, 20–100+ actuators	kHz	Achromatic, ~95%+	**Detection pupil**
Deformable lens	Continuous phase, transmissive	~100 Hz	Good	In-line detection MCAO
Piston micro-mirror arrays (UV-capable)	Phase	kHz	High, works to ~322 nm	Emerging
Etched silica / metasurface / DOE	Fixed complex	Static	Best available	Deployed PSF masks

The generalisation that survives all of this: **programmable phase belongs in the excitation pupil; programmable amplitude belongs in the detection image plane; fixed or low-order phase belongs in the detection pupil.** Every mature system in the tables above obeys that rule, and the ones that violate it (FINCH, SLM-based PSF engineering) pay for the violation in photons.

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6. Differentiable optics: three frameworks and the distance to metal

The previous section sorted *techniques* by build status. This section does the same for the *design methodology* itself — the differentiable, GPU-accelerated wave-optics frameworks that increasingly generate the designs. Three papers define the current state: FourierNets (Deb et al., NeurIPS 2022), XLuminA (Rodríguez, Arlt, Möckl & Krenn, Nat. Commun. 2024), and Chromatix (Deb, Both, … Turaga, Nat. Methods 2026).

They are frequently discussed as if they were a single trend. They are not. They sit at very different distances from a working instrument, and the pattern of *which* of their results have been realised in hardware is the most informative thing about them.

6.1 What each one is

	FourierNets	XLuminA	Chromatix
Object	An architecture + a large-scale differentiable simulator	A differentiable virtual optical table + topology search	A standard differentiable wave-optics library
Optimises	Pupil phase mask **and** decoder network, end to end	Optical *topology* plus element parameters	Whatever the user composes
Core claim	Deep decoders have a locality bias that blocks optimisation of highly non-local encoders; a shallow global-kernel Fourier-convolution decoder removes it	The space of optical layouts (~10^{20} discrete arrangements in their ansatz) can be searched by making topology continuous	Standardisation + GPU parallelism (2–6× on one GPU, up to 22× on eight)
Scale	Encoders 170×–7372× larger than prior end-to-end work	4–6.3 M optical parameters per run	Ring deconvolution over a FOV that overflows a single 80 GB H100
Hardware realised	**Partially** (see §6.2)	**None**	**Mixed, in three clean tiers** (see §6.3)

6.2 FourierNets: a prototype that measured a PSF and stopped

This is the design that motivated this project's own hardware study, so it is worth being exact about what exists.

The paper builds a 3D snapshot microscope: a phase mask in the pupil compresses a volume into one 2D camera frame, and a FourierNet decodes the volume. The target is whole-brain larval zebrafish at camera rates above 100 Hz — roughly two orders of magnitude faster than conventional scanning.

The optical parameters, from the paper's own derivation. Nyquist sampling of the image plane requires $\Delta x^* = \lambda/(2 \cdot NA)$. With NA = 0.8 and λ = 0.532 μm, Δx^* = 0.3325 μm. For a field of view L = 823 μm this gives $N^* = L/\Delta x^*$ = **2476 samples across the pupil**; they simulate the mask at 2560 × 2560. The paper is explicit that this is a constraint on the *phase mask*, independent of camera sampling.

The prototype that was built. A 16×/0.8 NA water-dipping objective (Nikon N16XLWD-PF), a 200 mm tube lens (Thorlabs TL200CLS2), a polarising beamsplitter (Thorlabs PBS251) transmitting horizontally polarised light onto a **Meadowlark P1920-532** phase-only LCoS SLM in a conjugate pupil, imaged onto a Hamamatsu Orca-Flash4.0 sCMOS. Total magnification 18.35×, object-space pixel 0.354 μm.

What was actually measured. The optical encoder — the PSF — was characterised with an *artificial* point source: a 532 nm collimated laser diode focused to a diffraction-limited spot by a second, higher-NA objective (60×/0.9), displaced in 5 µm steps over 250 µm, with 100 frames averaged and dark-subtracted at each plane. The measured PSF was compared with the simulated one and found to be a good *qualitative* match.

What was not done. No fluorescent sample was imaged through the instrument. No volume was reconstructed from real camera data. The ROI-specific encoder claims — the paper's most interesting result, that a mask optimised for one class of region outperforms masks optimised for others — were evaluated only in simulation, on one dataset. The authors state both limitations plainly: they did not focus on hardware implementation, and the ROI-specific benefits are simulation-only.

So the honest status is: **a phase mask designed in simulation was displayed on a real SLM and its PSF was measured. That is where the metal stops.**

6.3 The three-tier split inside Chromatix

Chromatix's six demonstrations sort themselves by how much hardware stands behind them, and the sorting is the most useful thing in the paper for anyone planning an instrument.

Tier 1 — real measurements, real hardware, no new optics required.

- *Ring deconvolution microscopy* (Kohli et al., Nat. Methods 2025). Real data: a UCLA Miniscope v3 with a GRIN lens and a 2.2 µm-pixel sensor, imaging rabbit liver under 550 nm incoherent illumination. The insight is that most imaging systems are rotationally equivariant, so common aberrations vary along the radius from the field centre up to a rotation. Calibration then scales linearly with sensor rows instead of quadratically, which makes deconvolution of *spatially varying* aberrations tractable. Chromatix's implementation recovers detail across the whole field where standard spatially invariant deconvolution recovers only the centre — and reconstructs a larger field than the original PyTorch code, which cannot fit the full camera FOV on one H100.
- *CoCoA* (Kang, Zhang, Yu & Ji, Nat. Mach. Intell. 2024). Real data: fixed Thy1-GFP mouse brain slices under a #1.5 coverslip tilted 3° to emulate a cranial window, 488 nm widefield excitation, 1.1 NA collection. An implicit neural representation of the sample and a Zernike representation of the aberration are estimated jointly and self-supervised, from a single 3D measurement — **no wavefront sensor, no deformable mirror, no guide star.** On beads with deliberately imposed aberrations, the Chromatix implementation recovers the applied Zernike coefficients to 3.56 nm RMS against 6.97 nm for the original.
- *Refractive-index microscopy* (Chowdhury et al., Optica 2019; Yang et al., Biomed. Opt. Express 2024 — a DMD plus microlens array as a switchable illumination-angle scanner). Real data: the tail of a 24-hpf zebrafish embryo, 660 nm coherent illumination at many angles, 0.37 NA, 168 images, reconstructed to 400 × 800 × 800 voxels via multislice propagation. 3–13× faster than the original, hours to minutes, and roughly 25 lines of differentiable simulation against about 107 in the original.

Tier 2 — the hardware exists somewhere, but this demonstration is in silico.

- *Holoscope snapshot PSF engineering.* The extended-data figure is explicitly labelled as matching the *in*

silico Holoscope performance of the FourierNets work. The ground truth is a confocal stack; the camera image is simulated. The contribution is a ~7× training speedup, reducing optimisation of a single PSF from weeks to days.

- *DeepCGH* (Eybposh et al., Opt. Express 2020). The method itself runs on real SLMs for optogenetics; the Chromatix comparison is a simulated benchmark of 2048 × 2048 holograms at three planes.

Tier 3 — pure simulation, never built.

- *Multicolour snapshot PSF.* One phase mask engineered so that 25 wavelengths from 400 to 650 nm produce distinguishable fringe patterns, decoded by a FourierNet into a hyperspectral cube from a single monochrome frame. Beautiful, and entirely virtual.
- *Computer-generated holography through scattering tissue.* A hologram optimised through a multislice model of a *C. elegans* head with refractive index deliberately scaled 5× to exaggerate scattering, restoring near-uniform axial stimulation. The paper calls it an in silico demonstration and does not claim otherwise.

6.4 The pattern

The tiers are not random. They line up with one variable:

> **Differentiable optics has been overwhelmingly successful where it corrects or decodes hardware that already exists, and overwhelmingly unrealised where it prescribes hardware that does not.**

Tier 1 works precisely because nobody had to build anything: a Miniscope, a widefield microscope, a DMD illumination scanner. The computational layer was pure addition. Tier 3 stays virtual because it requires a specific new optic *and* a calibration of that optic that nobody has. Tier 2 is the revealing middle — the optic exists, and the loop was still never closed.

XLuminA is the extreme case of the same phenomenon. It has no hardware at all, and its ansatz explains why: the large-scale search space is built from 6 light sources, 9 beam splitters, 24 "super-SLMs" (48 physical SLMs, each pair imprinting independent phase on orthogonal polarisation components), 24 wave plates and 6 high-NA objectives. That is a search *scaffold*, not a buildable instrument. The value is conceptual — the discovered solutions are meant to be read for their physical principle and then simplified by hand.

Two further caveats on XLuminA's headline result, which matter if one is tempted to treat "9.45% smaller spot than both references" as an imaging claim:

- **The objective is a beam-quality proxy, not an image quality metric.** The loss is the inverse density of detected intensity above a threshold — area divided by above-threshold intensity. There is no sample, no fluorophore photophysics beyond a Beer–Lambert stand-in for stimulated emission, no photon budget, no detection noise, no reconstruction. A tighter effective spot in this metric is not the same as a better image.
- **The robustness study is honest but narrow.** They inject uniform phase noise of ±0.01–0.1 rad on SLMs and wave plates, misalignment of ±0.01–0.1 mm, 1% beamsplitter transmit/reflect error, and show 8-bit discretisation and even ±0.1–1 rad / ±0.1–1 mm still gives usable imaging in the 4f rediscovery. That is real

evidence against fragility. It says nothing about throughput, polarisation loss, chromatic behaviour, or whether the speckle-like optimised phase patterns survive on a device with finite pitch and fill factor.

6.5 Why the simulated design stalls: the FourierNets encoder, costed

This project's earlier hardware analysis of exactly this optical scheme gives the concrete failure modes. They are worth stating alongside the paper's own numbers, because together they explain the Tier-2 gap in a way no amount of GPU speed addresses.

Space-bandwidth deficit — the binding constraint. The design requires $N^* = 2476$ samples across the pupil. The prototype's Meadowlark P1920-532 is 1920 × 1152. The short axis falls short by a factor of ~2.15. The device cannot render the mask the simulator optimised; it can only render a version covering roughly 370 μm of the intended 823 μm field. This is not a calibration error and no reconstruction network recovers it — the information was never modulated.

Photon budget — the invisible tax. The PBS-plus-LCoS chain gives roughly 43% end-to-end throughput. In a photon-starved snapshot experiment running above 100 Hz, that is a factor-2.3 hit to SNR which the simulation never sees, because the simulator applies Poisson noise to an assumed photon count rather than to the measured throughput of that specific beam path.

Modulation type — a theorem, not an engineering limit. Substituting a DMD (fast, cheap, broadband, polarisation-agnostic) fails for a reason that cannot be optimised away. A real-valued pupil — an amplitude mask, or a binary 0/π phase mask — makes the PSF at +z the exact mirror image of the PSF at –z; the correlation between PSF(+z) and the mirrored PSF(–z) is 1.000. A snapshot encoder whose entire function is to distinguish depths is destroyed by that degeneracy. Time-dithering (PWM) does not rescue it, because the camera integrates a *sum of intensities*, not the PSF of a time-averaged field. Holographic encoding of a complex pupil on a DMD (superpixel or Lee methods) does restore phase, at a cost of 16–64× in space-bandwidth and roughly 14× in photons — on a device that was already short of 2476 samples.

What is *not* the problem — phase depth. Four bits (16 levels) already give an encoder fidelity of 0.995 against continuous phase. The 256–4096 levels of an LCoS device are wasted on this task. That is a liberating result, because it admits the TI phase-only MEMS modulator class: 4-bit piston phase, polarisation-agnostic, kHz-rate, no LCoS crosstalk at Nyquist. The In-Vision DLPM980VIS (2048 × 1088, 10.8 μm pitch, 5 kHz, 420–680 nm) covers only ~362 μm of field on its short axis. Holoeye GAEA-2 (4160 × 2464, 3.74 μm pitch) is the only commodity device that nominally spans the full 823 μm.

The endgame. For a *frozen* encoder, an etched DOE or metasurface reaches ~96% efficiency against 43% for the SLM chain. Reprogrammability is worth its 2.2× photon cost only while the design space is still being searched.

The excitation arm, for completeness. Where a second modulator patterns the illumination, phase beats amplitude by roughly two orders of magnitude on sparse targets: ~85% of the energy delivered into the target pattern with a phase SLM running Gerchberg–Saxton, against ~0.5% for an amplitude mask, because phase redistributes light instead of absorbing it. Four-bit quantisation costs almost nothing here either (85.1% against

86.2% for continuous phase). Speckle — contrast ~0.20 on a single hologram — is cured by averaging K holograms within one camera exposure, which is exactly what a 5 kHz MEMS device makes possible: K = 16–50 brings speckle contrast to 0.08–0.09. The residual limitation is structural: a single phase modulator controls only intensity $|E|^2$ across the full space-bandwidth; independent amplitude *and* phase control costs half the space-bandwidth (double-phase encoding, or a two-modulator cascade — the same §10 argument in a different arm). For fluorescence excitation this is not a real constraint, since the phase in the sample is irrelevant.

6.6 What a hardware–aware design loop would add

None of the three frameworks currently treats device physics as a first-class constraint. A loop that did would include:

1. **Device-native parameterisation.** Optimise on the actual pixel grid, pitch, fill factor and quantisation of the target modulator, with quantisation-aware training (straight-through estimators), rather than optimising a 2560^2 continuous-phase ideal and rounding afterwards.
2. **Measured throughput inside the loss.** Carry the real transmission (≈0.43 for a PBS/LCoS chain, ≈0.96 for an etched DOE) and a Poisson model at the true expected photon count, so the optimiser sees the SNR consequences of its choices.
3. **Space-bandwidth as a hard constraint.** If N_device < N*, the objective's field of view must shrink explicitly rather than alias silently.
4. **Modulation type as a physics-carrying discrete choice.** Encode the defocus-sign degeneracy of real-valued pupils as a term in the loss so the optimiser cannot propose an unbuildable encoder in the first place.
5. **Hardware in the loop.** Replace the final stages of simulated propagation with the physical system. This is standard practice in display holography (camera-in-the-loop neural holography) and in hybrid SLM/DOE design for extended depth of field, and it is precisely what turns a "good qualitative match" between simulated and measured PSFs into a quantitative one.
6. **Real2sim fabrication models.** For etched or nanoprinted masks, a learned model of what the fabrication process will actually produce. Inverse-designed phase masks written by two-photon 3D nanoprinting with a two-step pre-compensation routine have delivered close agreement between predicted and measured performance for a ~4× depth-of-field extension in a real microscope — an existence proof that the design-to-fabrication loop can be closed.
7. **Self-calibrating reconstruction.** CoCoA-style joint estimation of sample and aberration means the device does not need to be perfect, only characterisable — which relaxes the tolerance stack considerably.

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7. Shifting complexity from optics to computation

The implicit thesis behind all three frameworks — and behind most of §4's digital-pinhole family — is that optical complexity can be traded for computational complexity at favourable exchange rates. That thesis is largely right, and it has hard limits. Both halves matter.

7.1 What genuinely moves

Optical function	Computational replacement	Evidence
Wavefront sensor + deformable mirror	Self-supervised joint sample/aberration estimation	CoCoA: Zernike coefficients to 3.56 nm RMS from a single 3D widefield measurement
Multi-conjugate AO for field-varying aberration	Ring deconvolution exploiting rotational equivariance	Full-FOV correction on a GRIN Miniscope with no added optics
Separate excitation and emission correctors	Dual deconvolution of the incoherent response matrix	130 nm at 180 μm depth in mouse brain; camera replaces the PMT (§3.8)
Confocal pinhole	Digital pinhole / pixel reassignment / Hadamard software aperture	1.53× transverse resolution with the photons a pinhole would have thrown away (§4.6)
Axial scanning stage	Snapshot encoding + learned decoder	Holoscope, DiffuserCam, light-field microscopy
Spectral splitting optics	Learned multicolour PSF on a monochrome sensor	25 wavelengths from one frame — in silico only
Precisely known illumination	Statistical reconstruction from unknown speckle	RIM: ~2× resolution, immune to excitation-path aberration (§3.7)

That list is substantial. It includes the pinhole, the wavefront sensor, the filter wheel, the z-stage, and multi-conjugate correction hardware.

7.2 What does not move, and never will

Four hard floors, in order of how often they get glossed over.

Photons. Every reconstruction is bounded by the information actually collected. Computation redistributes information; it does not create it. This bites hardest in snapshot modalities: multiplexing a volume onto one 2D frame means the shot noise of the *entire volume* contributes to the estimate of every voxel. The compression is free in acquisition time and expensive in SNR. Any claim that computation compensates for a lossy optical path is an accounting error.

The pupil cutoff, 2·NA/λ. No operation in the detection path — optical or computational — admits spatial frequencies the objective did not collect. Resolution beyond the cutoff requires the *illumination* to down-mix high frequencies into the passband (SIM, RIM, speckle saturation), or emitter sparsity and temporal statistics (SMLM, SOFI), or genuine prior knowledge. This is worth stating flatly because it is exactly where the "shift everything to computation" argument gets oversold. A better decoder does not enlarge the passband.

The space-bandwidth product of the modulator. §6.5 is the case study. If the device cannot render the mask, the encoding does not happen, and no decoder recovers it.

The forward-model gap. Every computational method is capped by the fidelity of its physical model. This is why bridging sim-to-real matters more than another order of magnitude of GPU throughput — and why Chromatix's most useful property may turn out to be composability with hardware control software rather than raw speed.

And one soft floor that deserves its own paragraph:

Prior-induced hallucination. A learned decoder with strong priors produces structure that is plausible rather than measured. For a discovery instrument this is a hazard, not a feature. The further the design shifts toward computation, the more of the budget must go to demonstrating that the output is measurement-driven: Fourier ring correlation, validation against an orthogonal modality with different failure modes, phantom and bead ground truth, per-voxel uncertainty, and held-out structures the network never saw. This is the cost that never appears in the speedup tables.

7.3 The rational division of labour

Given the above, the optics should retain responsibility for exactly four things:

1. **Collecting photons** — numerical aperture, transmission, detector quantum efficiency. Cheap in optics, impossible in software, and it sets the ceiling on everything else.
2. **Not destroying information** — avoid degeneracies (the real-pupil defocus-sign theorem is the canonical example), avoid aliasing, avoid saturation. An encoder that maps two distinct object states to the same measurement has made the inverse problem unsolvable, not merely hard.
3. **Conditioning the inverse problem** — this is what PSF engineering, structured illumination and snapshot encoders are actually *for*. The right question to ask of an optical element is no longer "does it form a sharp image" but "does it make the inverse problem well-conditioned at the photon budget available." That reframing is the real content of the differentiable-optics movement.
4. **Delivering excitation efficiently** — the ~85% versus ~0.5% gap between phase and amplitude patterning on sparse targets is a photon argument, and therefore an optics argument.

Everything else — inverting, correcting, sectioning, refocusing, unmixing, denoising — belongs in computation.

7.4 A ranked route to the best achievable image

1. **Fix the photon budget before anything else.** Etched DOE or metasurface over a programmable SLM once the design is frozen (≈96% vs ≈43%). Phase over amplitude in the excitation arm. Deformable mirror over LCoS in the detection pupil. These are multiplicative gains that no reconstruction recovers.
2. **Delete hardware that only corrects.** Wavefront sensors, physical pinholes, filter wheels, z-stages — replace with computational AO, digital pinholing, learned spectral encoding, snapshot capture. Each deletion also removes a transmission loss and an alignment degree of freedom.
3. **Keep programmable hardware only where it conditions the inverse problem.** In practice this is one modulator in the excitation pupil and one corrector in the detection pupil — the §11 architecture.
4. **Close the loop between hardware and model.** Hardware-in-the-loop mask refinement, self-calibrating reconstruction, real2sim fabrication models. This is where the largest unrealised gain currently sits: not in

better optics or better networks, but in the measurement that connects them.

5. **Validate against a modality with different failure modes** before believing any reconstruction that a prior helped produce.

7.5 A staged plan for the snapshot encoder in this project

Applying all of the above to the specific case — turning the FourierNets/Holoscope design into a working instrument — gives a concrete sequence:

Stage 0 — port, don't rebuild. Re-run the encoder optimisation in Chromatix rather than the original PyTorch code. Chromatix already contains a validated Holoscope implementation, matches the original reconstruction quality within statistical noise, and trains roughly 7× faster (weeks to days for a single PSF). This is the highest-leverage step available because the library, the forward model and the reference numbers all already exist.

Stage 1 — device-native, quantisation-aware optimisation. Re-optimise the mask on the actual device grid, with the real transmission and photon budget in the loss, for two candidates:

- *Holoeye GAEA-2* (4160 × 2464, 3.74 µm pitch) if the full 823 µm field is required — the only commodity device that nominally spans it.
- *In-Vision DLPM980VIS* (TI PLM, 2048 × 1088, 10.8 µm pitch, 5 kHz, 4-bit piston phase, polarisation-agnostic) if a ~362 µm field is acceptable in exchange for kHz reprogramming and no polarisation penalty. The 4-bit depth costs essentially nothing (encoder fidelity 0.995).

Expect the optimiser to return a *different* mask than the 2560^2 ideal — that is the point.

Stage 2 — hardware in the loop. Measure the PSF stack the way the original prototype did (artificial point source from a focused 532 nm diode, 5 µm steps over 250 µm, 100 frames averaged, dark-subtracted) but feed it back as the forward model instead of using it as a qualitative comparison figure. This is the single step the original work did not take, and it converts the encoder from "approximately as simulated" to "characterised."

Stage 3 — retrain the decoder on measured PSFs, then validate. A FourierNet trained on the simulated encoder will not be optimal for the measured one. Validate reconstructions against confocal or light-sheet ground truth on the same specimen, and test whether the ROI-specific encoder advantage — currently a simulation-only claim on a single dataset — survives real data. If it does not, the case for a programmable device over a fixed one weakens considerably.

Stage 4 — freeze and etch. Once the encoder stops changing, move to an etched DOE or metasurface for the production instrument and recover the ~2.2× in photons. Retain the SLM only if Stage 3 demonstrated a real ROI-specific benefit worth that cost.

Parallel track — the excitation arm. If patterned excitation is wanted, a phase modulator running Gerchberg–Saxton delivers ~85% of energy into a sparse target against ~0.5% for amplitude masking. At 5 kHz, averaging 16–50 holograms inside a single camera exposure suppresses speckle contrast from ~0.20 to 0.08–0.09. Full independent amplitude-and-phase control would cost half the space-bandwidth, and is unnecessary here

because the phase of the excitation field in the sample does not matter for fluorescence.

What XLuminA is for in this picture. Not this. XLuminA searches *topologies*; Chromatix optimises *parameters within a topology*. If the question is ever "is there a fundamentally different arrangement of elements that encodes depth better than a single pupil mask," XLuminA is the right tool — with the caveat that its objective must first be replaced by an imaging metric with a real photon budget, and its output read as a physical principle to be simplified rather than a wiring diagram to be built.

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8. Comparative summary

System	Exc. control	Det. control	Full-FOV?	Volumetric	Resolution	Status
AO-LLSM (2018)	SLM: lattice + AO	DM: AO	Tiled sub-volumes	Yes	~230 × 230 × 370 nm	Built, disseminated
MOSAIC (2026)	SLM/DM, all modes	DM, all modes	Tiled	Yes	~260 × 260 × 431 nm (LLS)	Built
PAM	DMD amplitude	Same DMD	Yes (amplitude only)	Sectioning	Confocal-class	Mature
FINCH / FINCHSCOPE	Conventional	SLM complex, self-interference	Yes (shift-invariant)	Yes, single hologram	~1.5× Rayleigh; 149 nm (crystal)	Built
COACH / I-COACH	Conventional	Coded phase mask	Yes	Yes, snapshot	~Rayleigh–1.5×	Built
PSF engineering (Tetrapod, DH, learned)	Conventional	SLM/mask pupil filter	Shift-invariant; field-calibrated	Yes	10–30 nm localisation	Mature
AO 2P-ISIM (2017)	DM shared	Emission rescan + SH sensor	Patch	Yes	176 × 729 nm @250 µm	Built
RIM / EDF-RIM	Speckle (uncontrolled)	Conventional	Yes	3D / projective	~2× (SIM-class)	Built, commercialised
Dual deconvolution IRM (2025)	Computational	Computational	7×7 patches	Yes	130 nm @180 µm	Published + code
MPAO (2017)	Segmented pupil, independent	—	450 × 450 µm, non-planar	Yes	Diffraction-limited 2P	Built
Aberration-space holography (2025)	Single hologram, per-site	—	**Yes, genuinely**	Yes	8× FOV, 12× volume	Built (not a microscope)

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9. What has *not* been built

To state the gap precisely: **no published instrument performs simultaneous, space-variant, complex (amplitude and phase) modulation of both the illumination and the detection field across a full field of view.**

The obstacles, in order of severity:

1. **Incoherence forbids the detection half.** Per-field-point complex control of emitted light would require a space-variant operator — mathematically a full transmission matrix, not a pupil filter. A single modulator plane cannot implement one. Only a *cascade* can (see §10).
2. **Photon budget.** Every additional element in the emission path costs signal. A two-plane LCoS cascade in detection could plausibly cost 70–90% of the fluorescence before it reaches the camera. For most live-cell experiments that is fatal.
3. **Chromatic dispersion.** LCoS diffractive operation is strongly wavelength-dependent; fluorescence emission bands are 30–80 nm wide. Achromatised SLM imaging exists but adds elements and loss.
4. **Degrees of freedom.** Full space-variant control over an N-resolvable-point field needs $O(N^2)$ degrees of freedom. A megapixel SLM gives 10^6; a 1000 × 1000-point field needs 10^{12}. Only heavy structure (low-rank aberration models, principal isoplanatic modes, layered MCAO) makes this tractable.
5. **Calibration and speed.** Measuring a space-variant response and updating it faster than biological motion is unsolved at scale.

The realistic near-term targets, in ascending difficulty:

- Aberration-space holography applied to the *excitation* arm of a light sheet or two-photon microscope — probably the highest-value, lowest-risk step available today.
- MCAO or deformable-lens correction in the *detection* arm for extended-FOV widefield/light-sheet.
- Cascaded modulation (MPLC) in detection — the only architecture with the right mathematical structure, but photon-starved.
- Full co-design: end-to-end differentiable optimisation of excitation pattern, detection pupil, and reconstruction network jointly.

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10. Cascaded modulation: the architecture nobody has applied to fluorescence detection

Multi-plane light conversion (MPLC) — a sequence of phase masks separated by free-space propagation — implements *arbitrary unitary transformations* between input and output spatial-mode bases. Diffraction between planes mixes spatially separated regions of the beam, which is exactly what a single plane cannot do. Recent work has compiled arbitrary M-mode unitaries with a number of phase planes scaling roughly linearly with M, and demonstrated programmable interferometers on up to 16 spatial pixel modes using a single SLM folded into 10 planes.

For imaging, this is the missing piece: an MPLC in the detection path is, in principle, a programmable space-variant complex operator — a physical implementation of an inverse transmission matrix. It could undo anisoplanatic aberration optically rather than computationally, over the whole field, in one shot.

Nobody has done it for fluorescence, for a good reason: an MPLC is unitary and lossless in principle but lossy in practice, and each pass off an LCoS costs polarisation and diffraction efficiency. A 10-plane cascade in an emission path is not currently viable. Reflective phase plates, metasurfaces, or a folded few-plane (2–3) cascade are the plausible routes.

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11. A concrete build proposal

If the goal is to actually construct a dual-path complex-modulation volumetric microscope with the largest corrected field currently achievable, here is a defensible architecture.

11.1 Optical layout

Excitation arm

- Multi-line laser combiner, AOTF for fast blanking and power control.
- Beam expander onto a fast phase-only LCoS SLM (e.g. 1920 × 1152, ≤10 ms response, or a ferroelectric binary device at kHz if SIM patterning dominates).
- SLM sits in a plane conjugate to the objective pupil. It performs three superposed jobs:
    1. Beam type (lattice / Bessel / Gaussian sheet / holographic spot array).
    2. SIM pattern phases and orientations, if running SIM.
    3. **Aberration-space correction**: per-site Zernike kernels summed into the single hologram, so that multiple field positions are simultaneously pre-compensated.
- Annular mask at a conjugate pupil (for lattice light-sheet) to block unwanted diffraction orders.
- Optional second modulator conjugate to the dominant aberrating layer (coverslip / tissue surface) for MCAO on the excitation side.

Detection arm

- High-NA water-dipping objective, tube lens.
- Deformable mirror (≥69 actuators, e.g. ALPAO / Boston Micromachines) at the pupil conjugate — achromatic, polarisation-insensitive, ≥95% efficiency. This carries the primary emission-path correction.
- A second, *specimen-conjugate* corrector for MCAO. A transmissive **deformable lens** is attractive here: compact, in-line, and demonstrated for large-FOV detection correction in widefield and light-sheet.
- Switchable 4f relay containing an SLM or fabricated phase mask for **PSF engineering** — inserted only for single-molecule/volumetric-localisation modes, bypassed otherwise so that normal imaging keeps its photons.

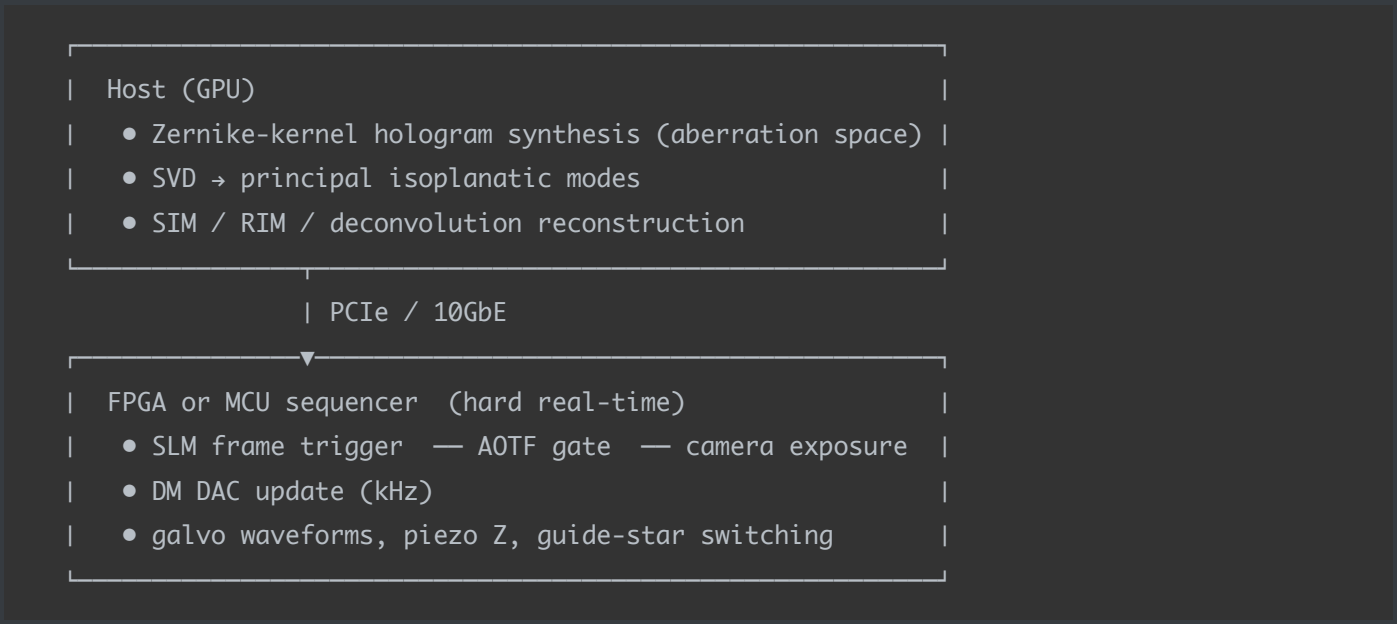
- Two sCMOS cameras via a dichroic for simultaneous two-colour.

Wavefront sensing

- Two-photon guide star, injectable through either arm via a switching galvanometer, as in AO-LLSM. Descanned onto a Shack–Hartmann sensor for direct sensing; fall back to sensorless modal optimisation (image sharpness metric, Zernike-mode dithering) where the guide star is impractical.
- Guide-star *scanning* over the target region to obtain an averaged correction — more robust than single-point, and it distributes photobleaching.

11.2 Control architecture

This is a hard real-time embedded problem as much as an optical one.



- All modulator updates, laser gating and camera exposures must be hardware-triggered from one clock. Software timing will not hold a SIM pattern sequence together.
- The DM loop runs fast (kHz DAC updates); the SLM loop runs at its device rate (10–100 Hz for nematic LCoS, kHz for ferroelectric binary).
- Hologram synthesis is the GPU-heavy part. Per-site Zernike kernels can be computed on the fly rather than cached: each GPU thread evaluates the Zernike basis for one SLM pixel, then builds all spot kernels from a small N × D coefficient matrix. This keeps memory bounded and allows video-rate updates.

11.3 Calibration procedure

1. **SLM/DM characterisation** — phase response vs. grey level per wavelength, flatness map, actuator influence matrix.
2. **Pupil registration** — map modulator pixels to pupil coordinates for both arms.
3. **Anisoplanatic wavefront mapping**, two complementary methods:

    - *Parallel superpixel interferometry*: diffract pairs of SLM superpixels to many field points at once and fit the fringes for relative phase. Gives amplitude at the modulator too, which good hologram synthesis needs. Parallelisation cuts measurement time by ~100×.
    - *Zernike subtraction*: sweep a global Zernike order, measure a per-site quality metric (spot size, or two-photon fluorescence brightness — which works without imaging), subtract the local contribution, iterate over orders.
4. **SVD compression** — decompose the measured coefficient matrix into principal isoplanatic modes and keep the first ~5–8. Beyond that, added modes mostly encode calibration noise.
5. **Emission-side PSF calibration on a field grid** — measure engineered PSFs at a lattice of field positions and axial planes, then interpolate at reconstruction time. This is how large-FOV 3D localisation is made to work with a field-varying PSF.

11.4 Modality set worth supporting

Mode	Excitation	Detection	Use case
LLS + dual AO	Lattice + AO	DM AO	Long-term gentle volumetric
LLS-SIM	Lattice, phase-stepped	DM AO	Super-resolved volumetric
3D-SIM	3-beam interference	DM AO	Thin-sample 2× in 3D
RIM	Speckle (SLM random masks)	DM AO	Deep, aberration-tolerant 2×
PSF-engineered SMLM	Widefield / oblique	Tetrapod or learned mask	3D localisation, large z-range
ISM / rescan	Multifocal	Rescan + DM	Confocal-class with 2× gain
2P point scanning	Aberration-space hologram, multifoci	Non-descanned	Deep in vivo

11.5 Realistic expectations

- Dual-path AO alone: 5–40× signal recovery and near-diffraction-limited resolution at 150–300 μm depth, per sub-volume.
- Aberration-space holography on the excitation arm: potentially 5–10× larger simultaneously corrected field than single-patch AO, on the excitation side only.
- MCAO or deformable lens on the detection arm: extended corrected detection field, best when aberration is dominated by one layer.
- Combined with SIM or RIM: ~2× beyond the corrected diffraction limit, i.e. roughly 100–150 nm lateral in favourable conditions.

- Photon budget is the binding constraint throughout. Every element added to the emission path must justify itself.

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12. Open questions worth working on

1. Can aberration-space holography be transplanted from projection to a *microscope excitation* arm, with a guide star supplying the per-site Zernike coefficients?
2. Is there a photon-viable few-plane cascade (2–3 reflective phase plates, or a metasurface stack) that gives useful space-variant correction in a detection path?
3. Can the incoherent-response-matrix dual deconvolution be run in closed loop, feeding measured excitation/emission aberrations back to physical correctors rather than only correcting in software?
4. End-to-end differentiable co-design of the excitation hologram, the detection pupil, *and* the reconstruction network, with the anisoplanatic forward model included — currently done for cameras and for single-molecule PSFs, not for whole dual-path microscopes.
5. Does the FINCH/COACH self-interference route have any headroom left at high NA, where photon efficiency has historically killed it?
6. Can a modern DMD-based programmable array microscope be revived as the detection front end for a dual-path AO instrument — using the non-conjugate (rejected) channel as a free, real-time, field-resolved wavefront-quality metric for sensorless correction across many isoplanatic patches at once?
7. What does a differentiable-optics framework look like when device physics is a first-class citizen — pixel pitch, quantisation, fill factor, measured transmission, and modulation-type degeneracies inside the loss rather than checked afterwards? None of the three current frameworks does this, and it is the difference between Tier 2 and Tier 1 in §6.3.
8. Would closing the hardware-in-the-loop on the FourierNets/Holoscope encoder — feeding measured PSF stacks back as the forward model rather than using them as a qualitative comparison — be enough to move 3D snapshot microscopy from a simulated result to a working instrument? This is the most testable open question in the document.
9. Does the ROI-specific encoder advantage survive on real data? It is currently a simulation-only claim on a single zebrafish dataset, and the entire case for a programmable rather than an etched pupil rests on it.
10. How should a computationally-heavy instrument prove that its output is measurement-driven rather than prior-driven? The validation methodology has not kept pace with the reconstruction methods.

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